



**HỌC VIỆN
NGÂN HÀNG**
BANKING ACADEMY OF VIETNAM

Web Design

Lecture 2 - HTML



2.1. Creating a simple page

2.2. HTML syntax

2.3. Marking up text

2.4. Adding links



2.1. Creating a simple page

Step 1

Start with content. As a starting point, we'll write up raw text content and see what browsers do with it.

Step 2

Give the document structure. You'll learn about HTML element syntax and the elements that set up areas for content and metadata.

Step 3

Identify text elements. You'll describe the content using the appropriate text elements and learn about the proper way to use HTML

Step 4

Add an image. By adding an image to the page, you'll learn about attributes and empty elements.

Step 5

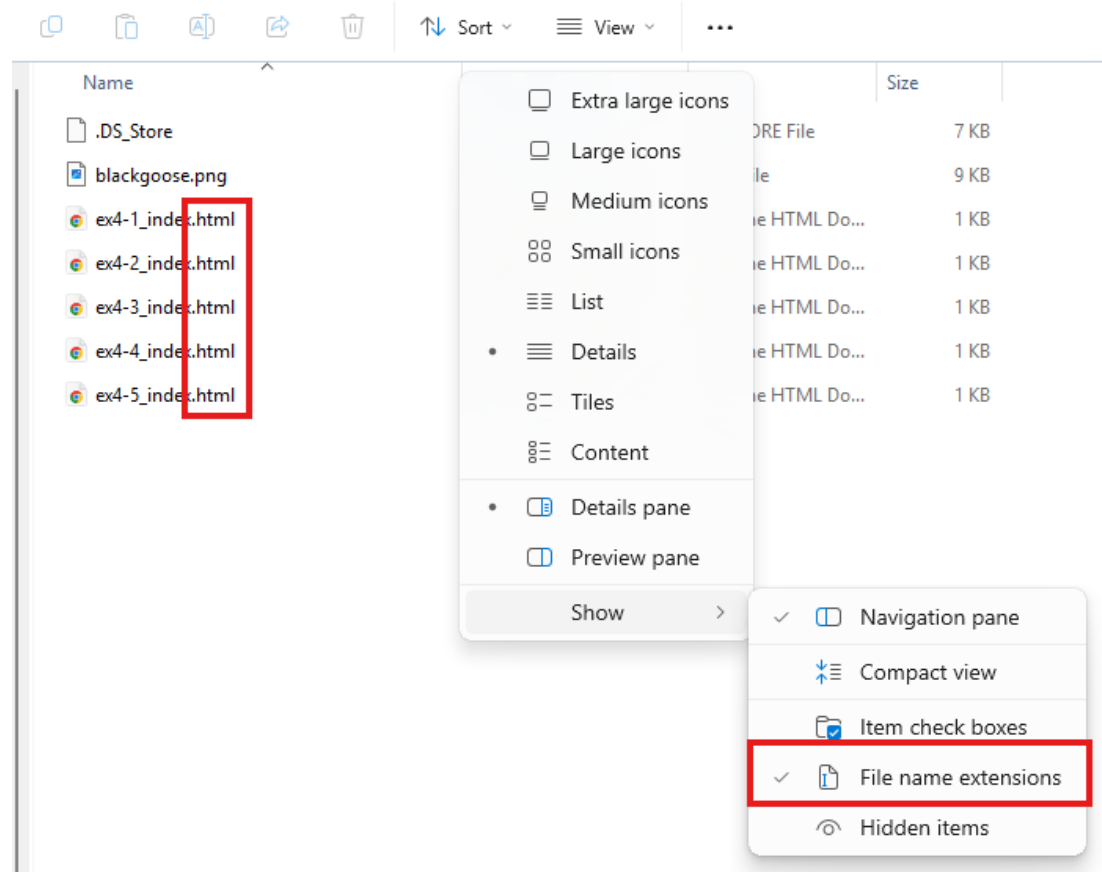
Change how the text looks with a style sheet. This exercise gives you a taste of formatting content with Cascading Style Sheets



Creating a New Document in :

- Notepad (Windows)
- TextEdit (MacOS)

Remember to show the file extension





Exercise 2-1 (3 mins): Type the home page content below into the new document in your text editor. Copy it exactly as you see it here, keeping the line breaks the same for the sake of playing along.

Select “Save” or “Save as” from the File menu, give it the name: ***index.html***

Black Goose Bistro

The Restaurant

The Black Goose Bistro offers casual lunch and dinner fare in a relaxed atmosphere. The menu changes regularly to highlight the freshest local ingredients.

Catering

You have fun. We'll handle the cooking. Black Goose Catering can handle events from snacks for a meetup to elegant corporate fundraisers.

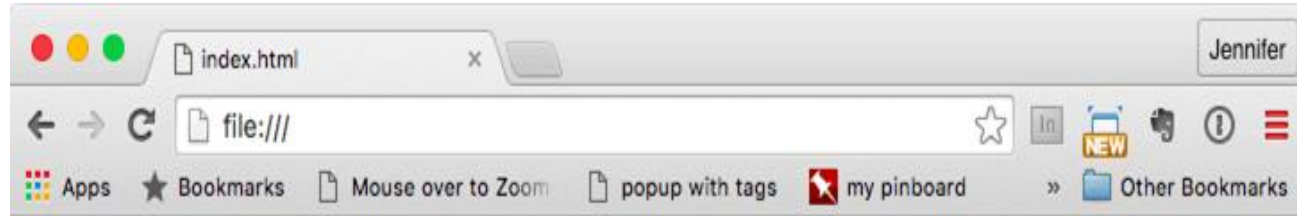
Location and Hours

Seekonk, Massachusetts;

Monday through Thursday 11am to 9pm; Friday and Saturday, 11am to midnight



You should see something like the page in the figure beside



Black Goose Bistro The Restaurant The Black Goose Bistro offers casual lunch and dinner fare in a relaxed atmosphere. The menu changes regularly to highlight the freshest local ingredients. Catering You have fun. We'll handle the cooking. Black Goose Catering can handle events from snacks for a meetup to elegant corporate fundraisers. Location and Hours Seekonk, Massachusetts; Monday through Thursday 11am to 9pm; Friday and Saturday, 11am to midnight



Use proper suffixes for your files. HTML files must end with **.html or .htm**. Web graphics must be labeled according to their file format: .gif, .png, .jpg (.jpeg is also acceptable, although less common), or .svg.

Never use character spaces within filenames. It is common to use an underline character or hyphen to visually separate words within filenames, such as robbins_bio.html or robbinsbio.html.

Avoid special characters such as ?, %, #, /, :, ;, •, etc. Limit filenames to letters, numbers, underscores, hyphens, and periods. It is also best to avoid international characters, such as the Swedish å.

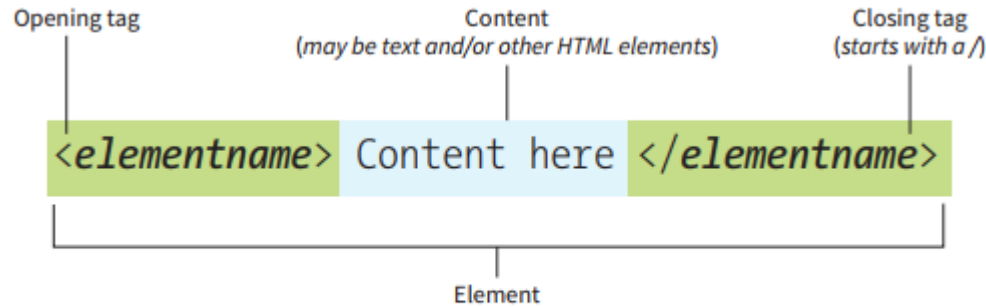


Filenames may be case-sensitive, depending on your server configuration. Consistently using all lowercase letters in filenames, although not required, is one way to make your filenames easier to manage.

Keep file names short. Long names are more likely to be misspelled, and short names shave a few extra bytes off the file size. If you really must give the file a long, multiword name, you can separate words with hyphens, such as a-long-document-title.html, to improve readability.



The Anatomy of an HTML Element



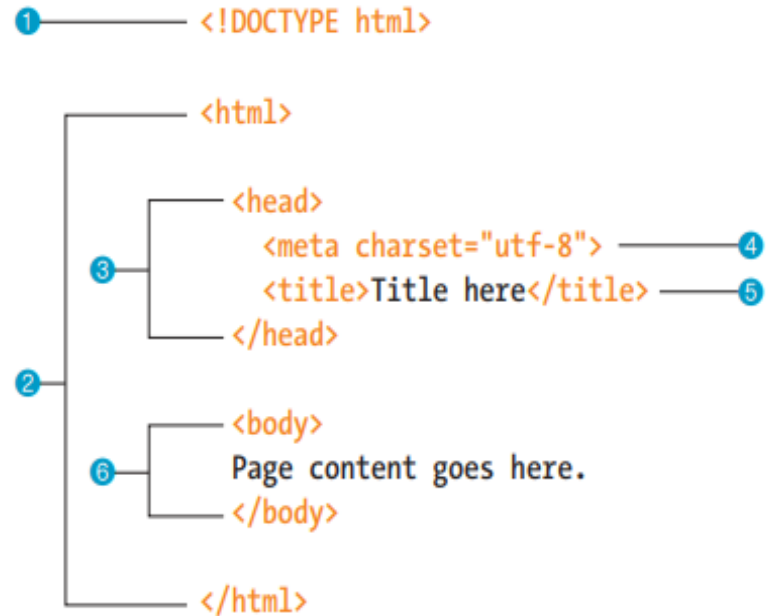
Example: `<h1>Black Goose Bistro</h1>`

Small tip: Slash Versus Backslash



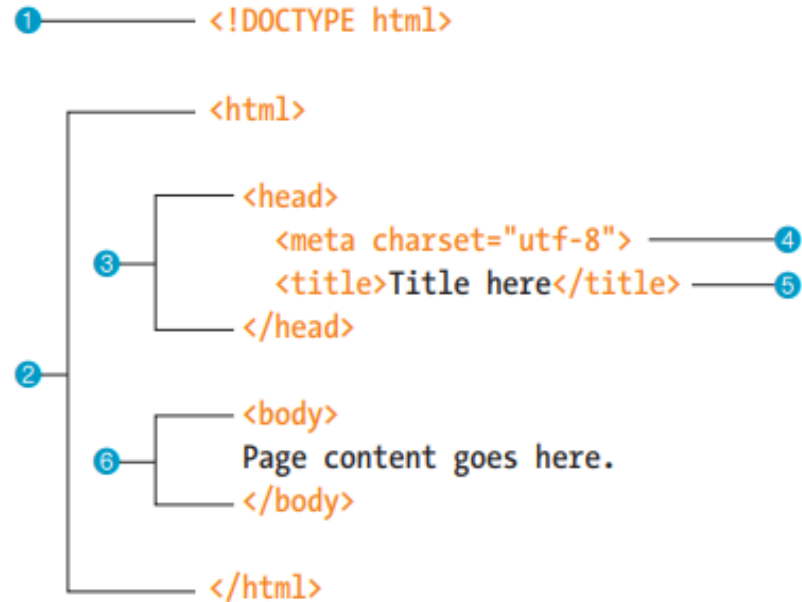


- **Document type declaration** (also called DOCTYPE declaration) that lets modern browsers know which HTML specification to use to interpret the document
- The entire document is contained within an **html** element. The html element is called the **root** element because it contains all the elements in the document, and it may not be contained within any other element.





- **Meta** elements provide document metadata, information about the document. In this case, it specifies the **character encoding** (a standardized collection of letters, numbers, and symbols) used in the document as Unicode version UTF-8 (see the sidebar “Introducing Unicode”). Other types of metadata provided by the meta element are the *author*, *keywords*, *publishing status*, and a *description* that can be used by search engines.



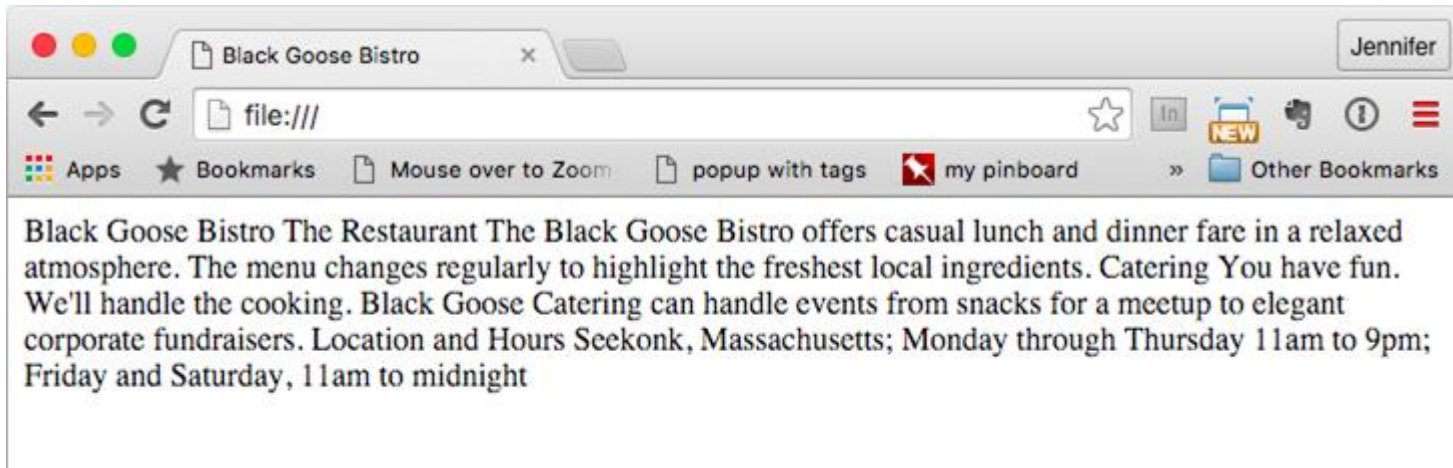
Exercise 2-2 (5 mins): Adding minimal structure



Open the new index.html document if it isn't open already

1. Add the **DOCTYPE** declaration
2. Put the entire document in an HTML root element by adding an **<html>** start tag after the DOCTYPE and an **</html>** end tag at the very end of the text
3. Next, create the document head that contains the title for the page. Insert **<head>** and **</head>** tags before the content. Within the head element, add information about the character encoding **<meta charset="utf-8">**, and the title, "Black Goose Bistro", surrounded by opening and closing **<title>** tags.
4. Finally, define the body of the document by wrapping the text content in **<body>** and **</body>** tags.
5. Save the document in the current directory, so that it overwrites the old version

Exercise 2-2 (5 mins): Adding minimal structure



Step 3: Identify text elements



Your job when marking up content is to choose the HTML element that provides the most meaningful description of the content. E.g. `<h1>`, `<p>`, `<b>`

In addition to adding meaning to content, the markup gives the document structure → **DOM**, for **Document Object Model**

Exercise 2-3 (5 mins): Defining text elements



1. Open the document **index.html** in your text editor, if it isn't open already.
2. The first line of text, "Black Goose Bistro," is the main heading for the page, so we'll mark it up as a Heading Level 1 (h1) element. Put the opening tag, `<h1>`, at the beginning of the line and the closing tag, `</h1>`, after it, like this:

`<h1>Black Goose Bistro</h1>`

3. Our page also has three subheads. Mark them up as Heading Level 2 (h2) elements in a similar manner. I'll do the first one here; you do the same for "Catering" and "Location and Hours."

`<h2>The Restaurant</h2>`

Exercise 2-3 (5 mins): Defining text elements



4. Each h2 element is followed by a brief paragraph of text, so let's mark those up as paragraph (p) elements in a similar manner. Here's the first one; you do the rest:

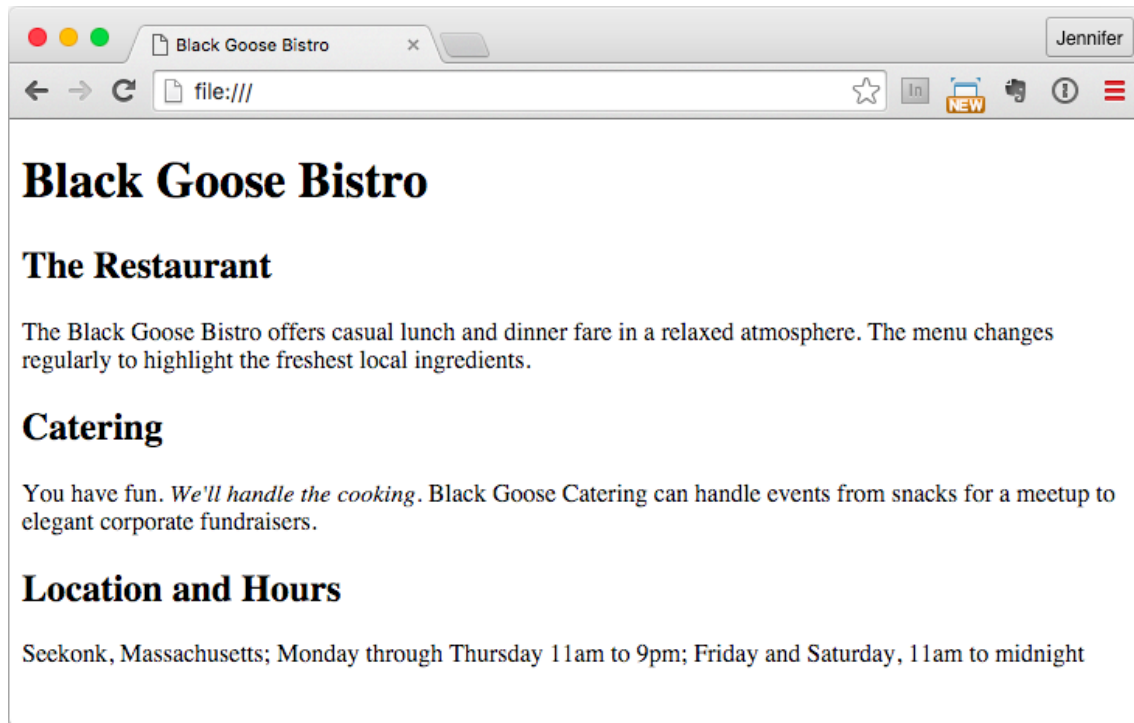
<p>The Black Goose Bistro offers casual lunch and dinner fare in a relaxed atmosphere. The menu changes regularly to highlight the freshest local ingredients.</p>

5. In the Catering section, I want to emphasize that visitors should just leave the cooking to us. To make text emphasized, mark it up in an emphasis element (em) element, as shown here:

<p>You have fun. We'll handle the cooking. Black Goose Catering can handle events from snacks for a meetup to elegant corporate fundraisers.</p>



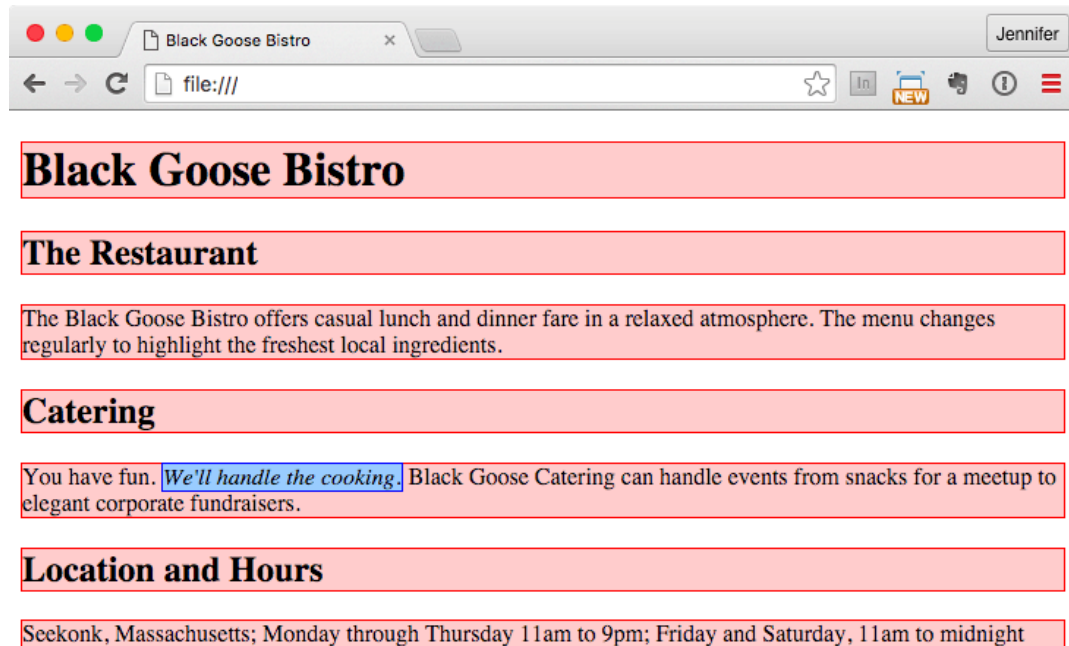
The result should be like this





By default, headings and paragraphs display as **block elements**. Browsers treat block elements as though they are in little rectangular boxes, stacked up in the page. Each block element begins on a new line.

Inline elements do not start new lines; they just go with the flow. E.g. the **** tag





How does the browser determine what an **h1** should look like?
It uses a style sheet! All browsers have their own built-in style sheets that describe the default rendering of elements

The default rendering is similar from browser to browser (for example, **h1s** are always big and bold), but there are some variations (the blockquote element for long quotes may or may not be indented).

How to custom the visual of these elements in our own way?

Step 4: Add an image



The image element (**img**) is an empty element. It tells the browser to get an image file from the server and insert it into the page.

```
<element-name>
```

Example: The `br` element inserts a line break.

```
<p>1005 Gravenstein Highway North<br>Sebastopol, CA 95472</p>
```



An **** tag is not very useful by itself—it doesn't indicate which image to use. That's where attributes come in.

Attributes are instructions that clarify or modify an element.

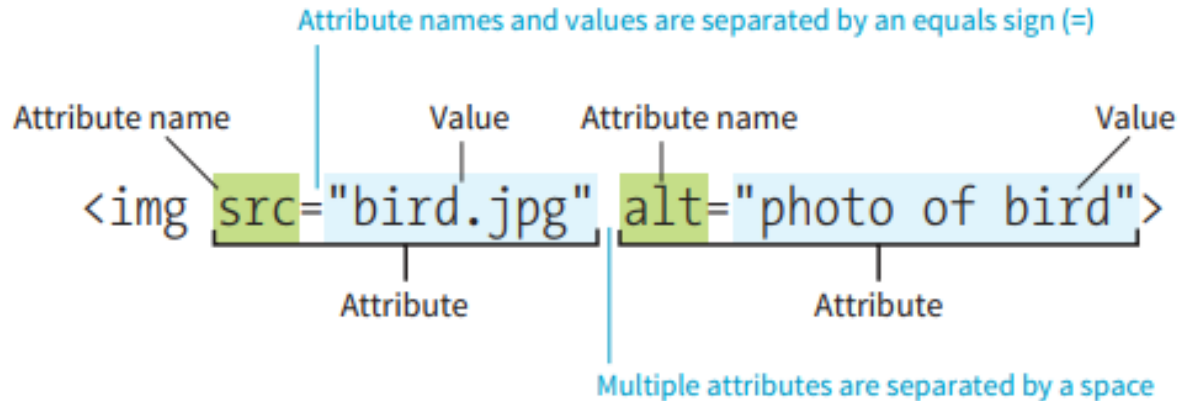
For the **img** element, the **src** (short for “source”) attribute is required, and specifies the location (URL) of the image file.



The syntax for an attribute is as follows:

`<element attributename="value">Content</element>`

`<element attribute1="value" attribute2="value">`



Exercise 2-4 (5 mins). Adding an image



1. Get a copy of the image file named ***blackgoose.png*** on your hard drive
2. Insert it at the beginning of the first-level heading by typing in the img element and its attributes as shown here:

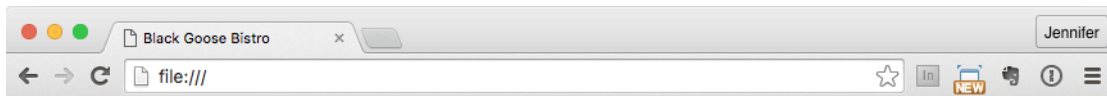
```
<h1>Black Goose  
Bistro</h1>
```

1. Add a line break (br) after the img element to start the headline text on a new line.

```
<h1><br>Black Goose  
Bistro</h1>
```



save ***index.html***
and open or
refresh it in the
browser window



Black Goose Bistro

The Restaurant

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Catering

You have fun. *We'll handle the cooking.* Black Goose Catering can handle events from snacks for a meetup to elegant corporate fundraisers.

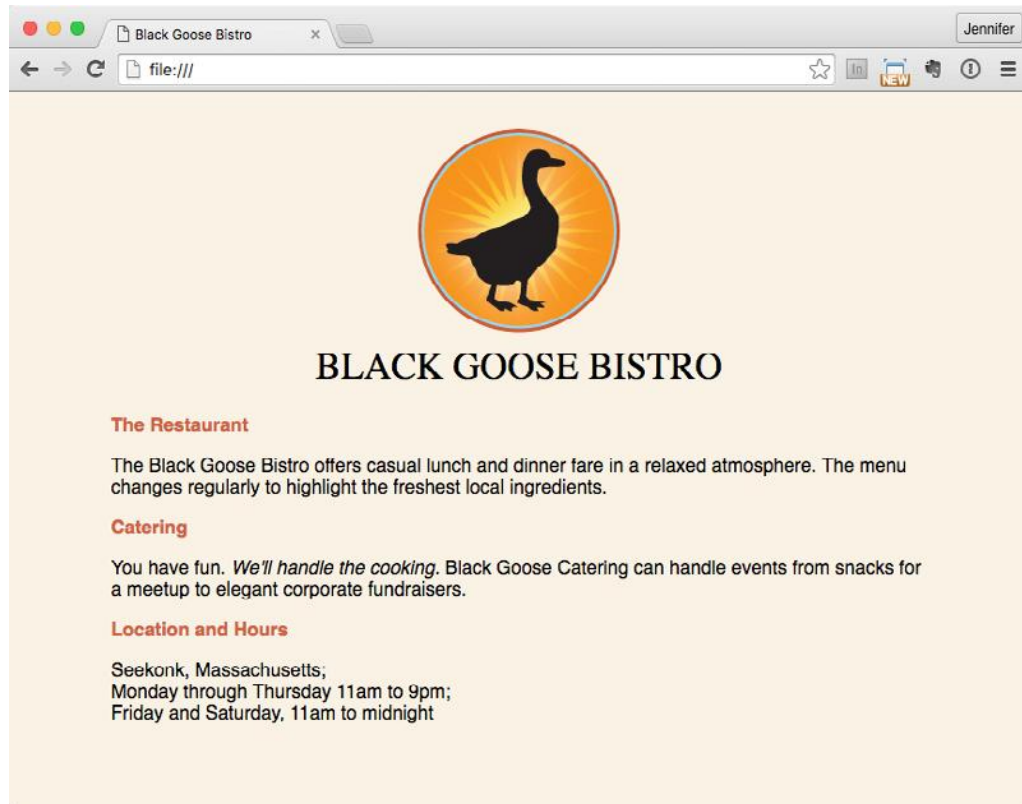
Location and Hours

Seekonk, Massachusetts;
Monday through Thursday 11am to 9pm;
Friday and Saturday, 11am to midnight



Change the appearance of the text elements and the page background by using some simple style sheet rules

→ Cascading Style Sheets (CSS)



Exercise 2-5 (3 mins). Adding a style sheet



The style element is placed inside the document head. Start by adding the style element to the document as shown here:

```
<head>
  <meta charset="utf-8">
  <title>Black Goose Bistro</title>
  <style>

  </style>
</head>
```

Exercise 2-5 (3 mins). Adding a style sheet



Copy the style code from the sample file to your own
index.html

Save the file and take a look at it in the browser.



All HTML documents must start with a document type declaration: **<!DOCTYPE html>**.

The HTML document itself begins with **<html>** and ends with **</html>**.

The visible part of the HTML document is between **<body>** and **</body>**.



The HTML element is everything from the start tag to the end tag:

<tagname>Content goes here...**</tagname>**

HTML elements can be nested (this means that elements can contain other elements).

HTML elements with no content are called empty elements.

HTML tags are not case sensitive: **<P>** means the same as **<p>**.



All HTML elements can have attributes

Attributes provide additional information about elements

Attributes are always specified in the start tag

Attributes usually come in name/value pairs like:

name="value"



Paragraphs

Headings

Thematic breaks

Lists

Others



Paragraphs are the most rudimentary elements of a text document. Indicate a paragraph with the p element by inserting an opening **<p>** tag at the beginning of the paragraph and a closing **</p>** tag after it

```
<p>Sans-serif fonts do not have serif slabs; their strokes are square  
on the end. Helvetica and Arial are examples of sans-serif fonts.  
In general, sans-serif fonts appear sleeker and more modern.</p>
```

Paragraphs may contain text, images, and other *inline* elements, but they **may not contain headings, lists, sectioning elements, or any elements that typically display as blocks by default**



When you add headings to content, the browser uses them to create a document outline for the page.

Assistive reading devices such as screen readers use the document outline to help users quickly scan and navigate through a page.

Search engines look at heading levels as part of their algorithms

h1 — **Type Design**

h2 — **Serif Typefaces**

Serif typefaces have small slabs at the ends of letter strokes. In general, serif fonts can make large amounts of text easier to read.

h3 — **Baskerville**

h4 — **Description**

Description of the Baskerville typeface.

h4 — **History**

The history of the Baskerville typeface.

h3 — **Georgia**

Description and history of the Georgia typeface.

h2 — **Sans-serif Typefaces**

Sans-serif typefaces do not have slabs at the ends of strokes.



If you want to indicate that one topic has completed and another one is beginning, you can insert what the spec calls a “*paragraph-level thematic break*” with the **hr** element. The hr element adds a logical divider between sections of a page or paragraphs without introducing a new heading level.

```
<h3>Times</h3>
<p>Description and history of the Times typeface.</p>
<hr>
<h3>Georgia</h3>
<p>Description and history of the Georgia typeface.</p>
```

Times

Description and history of the Times typeface.

Georgia

Description and history of the Georgia typeface.



Unordered lists

Collections of items that appear in no particular order

Ordered lists

Lists in which the sequence of the items is important

Description lists (rarely used)

Lists that consist of name and value pairs, including but not limited to terms and definitions



Unordered list in default style

```
<ul>  
  <li>Serif</li>  
  <li>Sans-serif</li>  
  <li>Script</li>  
  <li>Display</li>  
  <li>Dingbats</li>  
</ul>
```

- Serif
- Sans-serif
- Script
- Display
- Dingbats



Unordered list in
with appropriate
style sheets

Serif
Sans-serif
Script
Display
Dingbats

SERIF
SANS-SERIF
SCRIPT
DISPLAY
DINGBATS

Serif

Sans-serif

Script

Display

Dingbats

SERIF

SANS-SERIF

SCRIPT

DISPLAY

DINGBATS



Ordered list in default style

```
<ol>  
  <li>Gutenberg develops moveable type (1450s)</li>  
  <li>Linotype is introduced (1890s)</li>  
  <li>Photocomposition catches on (1950s)</li>  
  <li>Type goes digital (1980s)</li>  
</ol>
```

1. Gutenberg develops moveable type (1450s)
2. Linotype is introduced (1890s)
3. Photocomposition catches on (1950s)
4. Type goes digital (1980s)



Long Quotations

Preformatted Text

Figures

→ Self reading



Exercise 2-6. Marking up a recipe (15 mins)



Find the raw text of a recipe. It's up to you to decide which element is the best semantic match for each chunk of content. You'll use paragraphs, headings, lists, and at least one special content element.

Page 81

Tapenade (Olive Spread)

This is a really simple dish to prepare and it's always a big hit at parties. My father recommends:

"Make this the night before so that the flavors have time to blend. Just bring it up to room temperature before you serve it. In the winter, try serving it warm."

Ingredients

1 8oz. jar sundried tomatoes
2 large garlic cloves
2/3 c. kalamata olives
1 t. capers

Instructions

Combine tomatoes and garlic in a food processor. Blend until as smooth as possible.

Add capers and olives. Pulse the motor a few times until they are incorporated, but still retain some texture.

Serve on thin toast rounds with goat cheese and fresh basil garnish (optional).



The href attributes

Linking to pages on the web

Linking within your own site

Targeting a new browser window

Mail links



Absolute URLs:

href="http://www.oreilly.com/book1/recipes/index.html"

Relative URLs: href="recipes/index.html"

Opening anchor tag

```
<a href="https://www.amazon.com/Bequet-Gourmet-Caramel-24oz  
-Celtic/dp/B00GZEU10Y/ref=sr_1_1_a_it?ie=UTF8&qid=1467055107&s  
r=8-1&keywords=bequet">Bequet Caramels</a>
```

URL

Linked text

Closing anchor tag

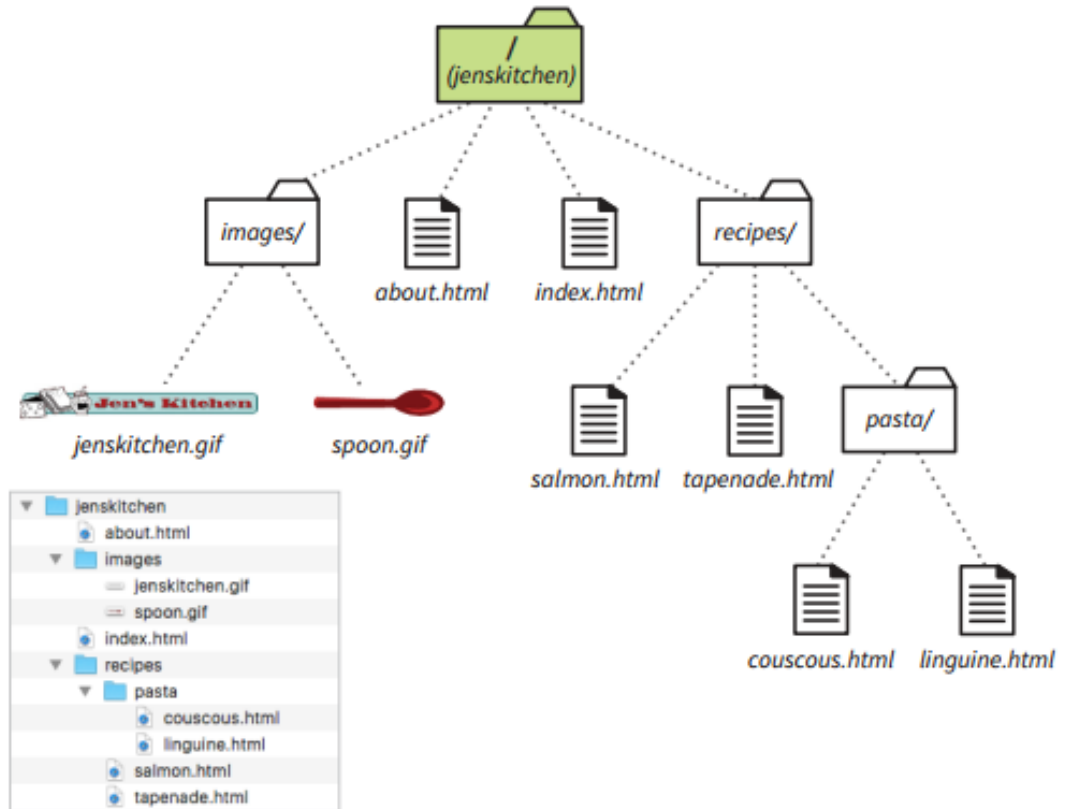


Most of the time, you'll want to create a link to a page on the web

```
<li><a href="http://www.foodnetwork.com">The Food  
Network</a></li>
```



A large portion of the linking you do is between pages of your own site: from the home page to section pages, from section pages to content pages, and so on. In these cases, you can use a **relative URL**





Linking Within a Directory

Linking to a Lower Directory

Linking to a Higher Directory

Linking with Site Root Relative Pathnames

Linking to a Specific Point in a Page



Setting ***target="_blank"*** always causes the browser to open a fresh window.

For example:

```
<a href="http://www.oreilly.com"  
target="_blank">O'Reilly</a>
```



A sample mailto link is shown here:

**`Contact Al
Klecker`**



Exercise 5-3 (page 110)

Test yourself (page 128): all 11 questions

Submit to the LMS, 1 contribution point for 2 first submissions (review by TAs)
